Unit 4: One-Sided Limits — Full Solutions

Warm-up

1. $f(x) = \begin{cases} x+1, & x<2 \\ 3x-1, & x \geq 2 \end{cases}$

For the left-hand limit, use the piece valid for $x<2$, which is $x+1$:

$$\lim_{x\to 2^-} f(x) = 2+1 = 3$$

For the right-hand limit, use the piece valid for $x \geq 2$, which is $3x-1$:

$$\lim_{x\to 2^+} f(x) = 3(2)-1 = 5$$

Since $3 \neq 5$, the two sides disagree.

Answer: left limit $=3$, right limit $=5$. The two-sided limit does NOT exist.

2. $f(x) = \begin{cases} x^2, & x<1 \\ 2x-1, & x \geq 1 \end{cases}$

Left-hand limit (use x^2): $1^2 = 1$.

Right-hand limit (use $2x-1$): $2(1)-1 = 1$.

Both sides give 1.

Answer: left limit $=1$, right limit $=1$. The two-sided limit exists and equals 1.

3. $f(x) = \begin{cases} 3, & x<0 \\ x+3, & x \geq 0 \end{cases}$

Left-hand limit (use the constant 3): 3.

Right-hand limit (use $x+3$): $0+3=3$.

Both sides give 3.

Answer: left limit $=3$, right limit $=3$. The two-sided limit exists and equals 3.

4. $f(x) = |x|$

For $x<0$: $|x| = -x$, so $\lim\limits_{x\to 0^-} f(x) = -0 = 0$.

For $x \geq 0$: $|x| = x$, so $\lim\limits_{x\to 0^+} f(x) = 0$.

Both sides give 0.

Answer: left limit $=0$, right limit $=0$. The two-sided limit exists and equals 0.

5. $f(x) = \begin{cases} 5-x, & x \leq 3 \\ x-1, & x>3 \end{cases}$

Left-hand limit (use $5-x$): $5-3=2$.

Right-hand limit (use $x-1$): $3-1=2$.

Both sides give 2.

Answer: left limit $=2$, right limit $=2$. The two-sided limit exists and equals 2.

6. $f(x)=\begin{cases} 2x, & x<-1 \\ x^2, & x\geq -1 \end{cases}$

Left-hand limit (use $2x$): $2(-1)=-2$.

Right-hand limit (use x^2): $(-1)^2=1$.

Since $-2\neq 1$, the two sides disagree.

Answer: left limit $=-2$, right limit $=1$. The two-sided limit does NOT exist.

Standard

7. $f(x)=\begin{cases} 1-x^2, & x\neq 1 \\ 2, & x=1 \end{cases}$

The piece $1-x^2$ applies for every x near 1 except exactly at 1 itself — so it's the correct formula to use approaching from both sides.

$$\lim_{x\to 1^-} f(x)=1-1^2=0, \qquad \lim_{x\to 1^+} f(x)=1-1^2=0$$

Both sides give 0.

Answer: left limit $=0$, right limit $=0$. The two-sided limit exists and equals 0. Notice this is different from $f(1)=2$ — but that's fine, since the limit never cares what the function's actual value is at the point itself.

8. $f(x)=\begin{cases} x^2-4, & x\neq 2 \\ 5, & x=2 \end{cases}$

The piece x^2-4 governs both sides near $x=2$:

$$\lim_{x\to 2^-} f(x)=2^2-4=0, \qquad \lim_{x\to 2^+} f(x)=2^2-4=0$$

Answer: left limit $=0$, right limit $=0$. The two-sided limit exists and equals 0 (again different from $f(2)=5$, which doesn't affect the limit).

9. $f(x)=\begin{cases} \dfrac{x^2-9}{x-3}, & x\neq 3 \\ 10, & x=3 \end{cases}$

Near $x=3$ (from either side), we use $\dfrac{x^2-9}{x-3}$. Factor the top: $x^2-9=(x-3)(x+3)$, so:

$$\frac{(x-3)(x+3)}{x-3} = x+3 \quad (\text{for } x \neq 3)$$

So both one-sided limits are:

$$\lim_{x\to 3}(x+3) = 3+3 = 6$$

Answer: $\lim_{x\to 3} f(x) = 6$. This is different from $f(3) = 10$ — the limit describes what the function approaches, not what it's actually defined to equal at that one point.

10. $f(x) = \begin{cases} x+2, & x < 0 \\ x^2+2, & x \geq 0 \end{cases}$

Left-hand limit (use $x+2$): $0+2 = 2$.

Right-hand limit (use x^2+2): $0^2+2 = 2$.

Answer: left limit $= 2$, right limit $= 2$. The two-sided limit exists and equals 2.

11. $f(x) = \begin{cases} 3x+2, & x \leq -2 \\ -x, & x > -2 \end{cases}$

Left-hand limit (use $3x+2$): $3(-2)+2 = -6+2 = -4$.

Right-hand limit (use $-x$): $-(-2) = 2$.

Since $-4 \neq 2$, the sides disagree.

Answer: left limit $= -4$, right limit $= 2$. The two-sided limit does NOT exist.

12. $f(x) = \dfrac{|x-3|}{x-3}$

For $x > 3$: $x-3 > 0$, so $|x-3| = x-3$, giving $f(x) = \dfrac{x-3}{x-3} = 1$.

$$\lim_{x\to 3^+} f(x) = 1$$

For $x < 3$: $x-3 < 0$, so $|x-3| = -(x-3)$, giving $f(x) = \dfrac{-(x-3)}{x-3} = -1$.

$$\lim_{x\to 3^-} f(x) = -1$$

Since $-1 \neq 1$, the sides disagree.

Answer: left limit $= -1$, right limit $= 1$. The two-sided limit does NOT exist.

Challenge

13. $f(x) = \begin{cases} x+3, & x < -1 \\ x^2-1, & -1 \le x < 2 \\ 2x-1, & x \ge 2 \end{cases}$

At $x = -1$:

Left-hand limit (use $x+3$, valid for $x < -1$): $-1+3 = 2$.

Right-hand limit (use x^2-1, valid starting at -1): $(-1)^2-1 = 1-1 = 0$.

Since $2 \neq 0$, the limit does not exist at $x = -1$.

At $x = 2$:

Left-hand limit (use x^2-1, valid up to but not including 2): $2^2-1 = 3$.

Right-hand limit (use $2x-1$, valid starting at 2): $2(2)-1 = 3$.

Since both sides give 3, the limit exists at $x = 2$ and equals 3.

Answer: at $x = -1$ — left limit $= 2$, right limit $= 0$, limit DOES NOT exist. At $x = 2$ — left limit $= 3$, right limit $= 3$, limit exists and equals 3.

14. $f(x) = \begin{cases} \dfrac{x^2-1}{x-1}, & x < 1 \\ 3, & x = 1 \\ 2x, & x > 1 \end{cases}$

Left-hand limit: for $x < 1$, simplify $\dfrac{x^2-1}{x-1} = \dfrac{(x-1)(x+1)}{x-1} = x+1$ (for $x \neq 1$).

$$\lim_{x \to 1^-} f(x) = 1+1 = 2$$

Right-hand limit (use $2x$): $2(1) = 2$.

Both sides give 2.

Answer: left limit $= 2$, right limit $= 2$. The two-sided limit exists and equals 2. This is different from $f(1) = 3$ — again, a totally normal and expected mismatch, since the limit doesn't depend on the function's defined value at the point.

15. $f(x) = \dfrac{x^2-4}{|x-2|}$

Factor the top for later use: $x^2-4 = (x-2)(x+2)$.

For $x > 2$: $|x-2| = x-2$, so

$$f(x) = \frac{(x-2)(x+2)}{x-2} = x+2$$

$$\lim_{x \to 2^+} f(x) = 2+2 = 4$$

For $x < 2$: $|x-2| = -(x-2)$, so

$$f(x) = \frac{(x-2)(x+2)}{-(x-2)} = -(x+2)$$

$$\lim_{x \to 2^-} f(x) = -(2+2) = -4$$

Since $-4 \neq 4$, the sides disagree.

Answer: left limit $= -4$, right limit $= 4$. The two-sided limit does NOT exist.

16. $f(x) = \begin{cases} kx+1, & x < 3 \\ x^2-2, & x \geq 3 \end{cases}$, find k so the limit exists at $x = 3$.

Left-hand limit (use $kx+1$): $k(3)+1 = 3k+1$.

Right-hand limit (use x^2-2): $3^2-2 = 7$.

For the limit to exist, the two sides must match:

$$3k+1 = 7$$

$$3k = 6$$

$$k = 2$$

Answer: $k = 2$.

17. $f(x) = \begin{cases} x^2+c, & x \leq 2 \\ 3x-4, & x > 2 \end{cases}$, find c so the limit exists at $x = 2$.

Left-hand limit (use x^2+c): $2^2+c = 4+c$.

Right-hand limit (use $3x-4$): $3(2)-4 = 2$.

Set them equal:

$$4+c = 2$$

$$c = -2$$

Answer: $c = -2$.

Applied

18. $T(x) = \begin{cases} 0.10x, & x \leq 50 \\ 0.12x - 1, & x > 50 \end{cases}$

Left-hand limit (use $0.10x$): $0.10(50) = 5$.

Right-hand limit (use $0.12x-1$): $0.12(50) - 1 = 6 - 1 = 5$.

Both sides give 5.

Answer: left limit $= 5$, right limit $= 5$. The limit exists and equals 5 — meaning there's no sudden jump in tax owed right at the $x = 50$ bracket boundary in this model; the tax owed transitions smoothly across the boundary.

19. $C(w) = \begin{cases} 5, & w \leq 2 \\ 5 + 2(w-2), & w > 2 \end{cases}$

Left-hand limit (use the constant 5): 5.

Right-hand limit (use $5 + 2(w-2)$): $5 + 2(2-2) = 5 + 0 = 5$.

Both sides give 5.

Answer: left limit $= 5$, right limit $= 5$. The limit exists and equals 5 — there's no price jump right at the 2-pound mark.

20. $P(h) = \begin{cases} 5, & 0 < h \leq 1 \\ 8, & 1 < h \leq 2 \\ 11, & 2 < h \leq 3 \end{cases}$

Left-hand limit at $h = 1$ (use the piece valid for $h \leq 1$, the constant 5): 5.

Right-hand limit at $h = 1$ (use the piece valid just above 1, the constant 8): 8.

Since $5 \neq 8$, the sides disagree.

Answer: left limit $= 5$, right limit $= 8$. The limit does NOT exist at $h = 1$. This tells you that the moment you cross from 1 hour into just over 1 hour, the price suddenly jumps from \$5 to \$8 — a real, sudden jump, unlike the smooth tax and shipping examples above.

21. $D(x) = \begin{cases} 30, & x \leq 2 \\ 30 + 10(x-2), & x > 2 \end{cases}$

Left-hand limit (use the constant 30): 30.

Right-hand limit (use $30 + 10(x-2)$): $30 + 10(2-2) = 30 + 0 = 30$.

Both sides give 30.

Answer: left limit $= 30$, right limit $= 30$. The limit exists and equals 30 — there's no sudden jump in cost right at the 2 GB mark; the overage charge phases in smoothly from that point.